NLP – Sentiment Analysis

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NLP Techniques

Input clean and verified data

Techniques advanced analysis techniques for text data

Output additional input about figuring out “what movies are similar to Spiderman: No Way Home”

NLP Techniques

Sentiment Analysis

Topic Modeling

Text Generation

Sentiment Analysis

input : A corpus. The reason we're not using the document-term matrix here is because order matters. "great" = positive. "not great" = negative

TextBlob : TextBlob is a Python library build on top of NLTK. It's easier to use and provides some additional functionality, such as rule-based sentiment scores.

output : Each movie will be given a sentiment score and subjectivity score.

Sentiment Analysis

```
# The code for TextBlob’s sentiment analysis
from textblob import TextBlob
Textblob(“I love Indonesia”).sentiment
```

```
Sentiment(polarity=0.5, subjectivity=0.6)
```

polarity

-1 ⟷ **+1**

subjectivity

0 ⟷ **+1**

How does textblob work?

[redacted]

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```
<word form="abhorrent" wordnet_id="a-1625063" pos="JJ" sense="offensive to the mind" polarity="-0.7" subjectivity="0.8" intensity="1.0" r
<word form="able" cornetto_synset_id="n_a-534450" wordnet_id="a-01017439" pos="JJ" sense="having a strong healthy body" polarity="0.5" su
<word form="able" wordnet_id="a-00001740" pos="JJ" sense="(usually followed by 'to') having the necessary means or skill or know-how or a
<word form="able" wordnet_id="a-00306663" pos="JJ" sense="having inherent physical or mental ability or capacity" polarity="0.5" subjecti
<word form="able" wordnet_id="a-00510348" pos="JJ" sense="have the skills and qualifications to do things well" polarity="0.5" subjectivi
<word form="above" cornetto_synset_id="n_a-504850" wordnet_id="a-00125993" pos="JJ" sense="appearing earlier in the same text" polarity="
<word form="abridged" cornetto_synset_id="d_a-9176" wordnet_id="a-00004413" pos="JJ" sense="(used of texts) shortened by condensing or re
<word form="abrupt" cornetto_synset_id="n_a-505100" wordnet_id="a-00640520" pos="JJ" sense="surprisingly and unceremoniously brusque in m
<word form="abrupt" cornetto_synset_id="n_a-529169" wordnet_id="a-01145151" pos="JJ" sense="extremely steep" polarity="0.0" subjectivity=
<word form="abrupt" wordnet_id="a-01143585" pos="JJ" sense="exceedingly sudden and unexpected" polarity="0.0" subjectivity="1.0" intensit
<word form="abrupt" wordnet_id="a-02294122" pos="JJ" sense="marked by sudden changes in subject and sharp transitions" polarity="0.0" sub
```

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How does textblob work?

Word Form	WordNet ID	POS	Sense	Polarity	Subjectivity
great	a-01123879	JJ	very good	1.0	1.0
great	a-012378818	JJ	of major significance or importance	1.0	1.0
great	a-01386883	JJ	relatively large in size or number or extent	0.4	0.2
great	a-01677433	JJ	remarkable or out of the ordinary in degree or magnitude effect	0.8	0.8

Sentiment Analysis

```
Textblob(“I love Indonesia”).sentiment
```

```
Textblob(“I love Indonesia”).sentiment
```

```
Textblob(“I love Indonesia”).sentiment
```

```
Textblob(“I love Indonesia”).sentiment
```

```
Sentiment(polarity=0.8, subjectivity=0.75)
```

```
Sentiment(polarity=1, subjectivity=0.75)
```

polarity x-0.5

subjectivity x 1.3

```
Sentiment(polarity=0.3, subjectivity=0.975)
```

TextBlob Sentiment

TextBlob finds all of the words and phrases that it can assign a polarity and subjectivity to, and averages all of them together

Output: each movie is assigned one polarity and one subjectivity score

While not the most sophisticated technique, it’s a good starting point

This rule-based implementation is a knowledge-based technique. There are also statistical methods out there such as Naive Bayes

Google Colab Code